1.Branch Instruction Verification

```
========================================
      BRANCH REGRESSION RESULTS
========================================
PASS: BEQ taken
PASS: BEQ not taken
PASS: BNE taken
PASS: BNE not taken
PASS: BLT taken
PASS: BLT not taken
PASS: BGE taken
PASS: BGE not taken
PASS: BLTU taken
PASS: BLTU not taken
PASS: BGEU taken
PASS: BGEU not taken

========================================
Tests passed : 12
Tests failed : 0
Total tests  : 12
RESULT: ALL BRANCH TESTS PASSED
========================================
```

2.Final ALU and Edge Tests

```
=== GROUP 1: ALU BOUNDARIES ===
PASS: ADD signed overflow = 80000000
PASS: ADD wraparound = 00000000
PASS: SUB underflow = ffffffff
PASS: SUB boundary = 7fffffff
PASS: AND = 00f00a50
PASS: OR = fff0aff5
PASS: XOR = ffffffff
PASS: SLT signed = 00000001
PASS: SLTU unsigned = 00000000

=== GROUP 2: SHIFT BOUNDARIES ===
PASS: SLL shift 0 = 00000001
PASS: SLL shift 31 = 80000000
PASS: SRL shift 31 = 00000001
PASS: SRA shift 31 = ffffffff
PASS: SRA shift 0 = ffffffff
PASS: Shift amount uses low five bits = 00000001
```

```
=== GROUP 3: ALU CONTROL ===
PASS: Address ADD = 0000
PASS: Branch SUB = 0001
PASS: R-type ADD = 0000
PASS: R-type SUB = 0001
PASS: ADDI with bit 30 set = 0000
PASS: SLL = 0111
PASS: SLT = 0101
PASS: SLTU = 0110
PASS: XOR = 0100
PASS: SRL = 1000
PASS: SRA = 1001
PASS: SRAI = 1001
PASS: OR = 0011
PASS: AND = 0010

=== FINAL RESULT ===
ALL ALU TESTS PASSED
```

3.CPU Integration Regression

```
===== CPU INTEGRATION REGRESSION =====
PASS: ADDI x1
PASS: ADDI x2
PASS: ADDI x3
PASS: LB offset 1
PASS: LH offset 2
PASS: LW after SB and SH
PASS: LBU offset 3
PASS: LHU offset 2
PASS: BEQ taken
PASS: BNE not taken
PASS: BNE taken
PASS: Negative immediate
PASS: BLT signed taken
PASS: BGE signed taken
PASS: BLTU unsigned not taken
PASS: Following BLTU
PASS: BGEU unsigned taken
PASS: JAL link address
PASS: JAL skips instruction
PASS: ADDI maximum positive
PASS: ADDI minimum negative
PASS: AUIPC at address 98
PASS: LUI
PASS: x0 read through ADDI
PASS: JALR target base
PASS: JALR link address
PASS: JALR target instruction
PASS: SB/SH byte lanes
PASS: Unused memory word
PASS: CPU architectural x0 read
PASS: Reached terminal JAL loop

===== IMMEDIATE GENERATOR BOUNDARIES =====
PASS: IMM I-type minimum -2048
PASS: IMM I-type maximum +2047
PASS: IMM S-type minimum -2048
PASS: IMM B-type minimum -4096
PASS: IMM B-type maximum +4094
PASS: IMM U-type sign bit
PASS: IMM J-type minimum -1048576

===== REGISTER FILE x0 TEST =====
PASS: x0 reads zero after attempted write

========================================
ALL REGRESSION CHECKS PASSED
```

4.Immediate Generator and Register File Verification

```
=== GROUP 1: REGISTER FILE ===
PASS: Write to x0 ignored
PASS: x0 reads zero on both ports

=== GROUP 2: IMMEDIATES ===
PASS: I-type minimum = fffff800
PASS: I-type maximum = 000007ff
PASS: S-type minimum = fffff800
PASS: S-type maximum = 000007ff
PASS: B-type minimum = fffff000
PASS: B-type maximum = 00000ffe
PASS: J-type minimum = fff00000
PASS: J-type maximum = 000ffffe
PASS: U-type high bit = 80000000
PASS: U-type maximum positive = 7ffff000

=== GROUP 3: CONTROL SAFETY ===
PASS: Unsupported load funct3
PASS: Unsupported store funct3
PASS: Unsupported opcode
PASS: Invalid store type preserves memory

================================
ALL FINAL CHECKS PASSED
================================
```

--

5.ALU and Memory Load Edge Tests

```
========================================
        FINAL ALU EDGE TESTS
========================================
PASS: ALU ADD
PASS: ALU SUB
PASS: ALU AND
PASS: ALU OR
PASS: ALU XOR
PASS: ALU SLT signed negative less than positive
PASS: ALU SLTU unsigned comparison
PASS: ALU SLL
PASS: ALU SRL
PASS: ALU SRA negative value
PASS: ALU SLL shift by zero
PASS: ALU SLL shift by 31
PASS: ALU SRL shift by zero
PASS: ALU SRL shift by 31
PASS: ALU SRA shift by 31
PASS: ALU SRA positive value by 31
PASS: CONTROL SLTIU decodes to SLTU
PASS: CONTROL SRLI decodes to SRL
PASS: CONTROL SRAI decodes to SRA
PASS: ALU SLTIU result
PASS: ALU SRLI result
PASS: ALU SRAI result
```

```
===== MEMORY LOAD EDGE TESTS =====
PASS: MEMORY LBU offset 0
PASS: MEMORY LBU offset 1
PASS: MEMORY LBU offset 2
PASS: MEMORY LBU offset 3
PASS: MEMORY LHU offset 0
PASS: MEMORY LHU offset 2
PASS: MEMORY LB signed offset 3
PASS: MEMORY LH signed offset 2

========================================
        FINAL REGRESSION SUMMARY
========================================
Tests passed : 30
Tests failed : 0
Total tests  : 30
RESULT: ALL MODULE EDGE TESTS PASSED
========================================
```

6.Final Regression Summary

```
=== TEST 1: AUIPC / IMMEDIATES ===
PASS: ADDI baseline = 00000005
PASS: AUIPC at PC 4 = 00000004
PASS: AUIPC at PC 8 = 00001008
PASS: ADDI negative immediate = ffffffff

=== TEST 2: x0 / NOP ===
PASS: x0 remains zero = 00000000
PASS: x0 read path returns zero

=== TEST 3: UNSUPPORTED OPCODE ===
PASS: Unsupported instruction preserves x1 = 00000005
PASS: SW executed; memory = 00000005
PASS: Final PC = 00000020

=== FINAL RESULT ===
ALL CHECKED RESULTS PASSED
```
